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# Guarded Block Transfers for Polytope Distance: Linear Convergence with Explicit Constants and Gap-Certified Screening for SVM Training

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## Abstract

Hard-margin SVM training is the distance problem between two convex hulls, and the Triangle Algorithm (TA) solves it with a certified duality gap from supporting hyperplanes. We study the weight-transfer steps used by the enhanced TA and propose *guarded block transfers*:  $k$  Mitchell–Demyanov–Malozemov (MDM) transfers per scan are aggregated into a single exact line search, compared against the single best transfer at a cost independent of  $n$ , and replaced by an away step whenever the leading transfer is capacity-clipped. Analysing the method on the Minkowski-difference polytope, where the objective is 1-strongly convex and one pyramidal width governs the geometry, we prove global linear convergence with explicit constants and without the combinatorial swap-step bounds of pairwise Frank–Wolfe. Because the TA’s certified gap is the Frank–Wolfe gap over that polytope divided by the current distance, gap-safe screening applies, and we give an explicit activation theorem: screening removes every zero-weight non-support point after  $O\left(\frac{k}{\rho} \log \frac{1}{\tau}\right)$  iterations, where  $\rho$  is the linear rate constant and  $\tau$  the optimal-face score margin. Every inequality in the analysis has a numerical check. On five LIBSVM benchmarks the solver matches the baselines’ solutions to five digits wherever both converge. It wins or ties on sparse-support cells (gisette’s linear, kernel and  $L_2$  margins; kernel  $L_2$  margins with expensive columns) against LIBSVM and kernel sequential minimal optimisation (SMO), and it is one to two orders of magnitude slower on dense-support  $L_2$  margins and nearly touching kernel hulls, where LIBLINEAR finishes in seconds. We report both sides.

## 1 Introduction

Training a linear hard-margin SVM is equivalent to finding the closest points of the two classes’ convex hulls: the optimal separator is the perpendicular bisector of the witness pair, and the margin is half the hull distance [8]. Kalantari’s Triangle Algorithm (TA) exploits this *distance duality* directly [1]. It comes in two forms: TA I decides whether the hulls intersect, and TA II computes their distance, maintaining an iterate pair  $(p, q)$  and a certified interval  $[LB, UB]$  on the distance from parallel supporting hyperplanes, so that it can stop at a provable relative accuracy  $\varepsilon$ . The enhanced TA (ETA) of [2] adds joint segment updates, dot-product caching, an anti-zig-zag heuristic and prioritised search, and reports large speedups over SMO on synthetic data.

The workhorse step of such geometric solvers is the *weight transfer*: move mass from the worst active vertex to the best candidate vertex, the Mitchell–Demyanov–Malozemov (MDM) step [4], which is exactly the pairwise Frank–Wolfe (FW) step of [3] on the relevant polytope; MDM converges linearly [24], and FW methods have a long history in SVM training [20, 22, 23]. Each step

costs an  $O(n)$  scan and produces one transfer; we ask how much of the scan can be reused across several transfers without weakening the guarantee.

## Contributions.

1. **Guarded block transfers** (§3): pair the top- $k$  receivers with the worst- $k$  active donors, take one exact line search along the aggregated direction, and *guard* by a comparison with the single MDM step that costs  $O(k^2)$  cached dot products, independent of  $n$ ; on capacity-clipped iterations, fall back to an away step. Cache updates for a whole block are one matrix–vector product.
2. **Linear convergence with explicit constants** (Theorem 1): on the Minkowski-difference polytope  $Z = \text{conv}(V) - \text{conv}(W)$  the guarded block algorithm inherits the pyramidal-width rate of pairwise FW up to a constant, and the away fallback replaces pairwise FW’s combinatorial swap-step bound with linear drop-step counting, as blended pairwise conditional gradients do with a different guard [12].
3. **Screening activation with explicit constants** (Theorem 2): the TA certificate is the FW gap divided by the distance (Lemma 3, an observation that goes back to Gilbert and Wolfe), so gap-safe screening [16] applies, and we bound the iteration by which it has removed every zero-weight non-support point in terms of the rate constant and the optimal-face margin.
4. **A regime study on real data** (§5). The analysis predicts  $O(n^2)$ -type cost when the optimal support is dense (pyramidal width  $\sim 1/\sqrt{n}$ , rate constant  $\sim 1/n$ ). Two batteries on five LIBSVM datasets show the boundary from both sides: the solver wins or ties on sparse-support high-dimensional cells and loses by orders of magnitude on dense-support  $L_2$  soft margins and on near-touching kernel hulls.

Code, proofs, numerical checks and experiment scripts are in the supplementary repository (anonymised for review).

## 2 Setting

Let  $V = \{v_1, \dots, v_n\}$ ,  $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$ . The polytope distance problem is

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} \frac{1}{2} \|p - q\|^2. \quad (1)$$

Writing  $z = p - q$  and  $Z := \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$ , (1) is the minimum-norm-point problem  $\min_{z \in Z} F(z)$ ,  $F(z) = \frac{1}{2} \|z\|^2$ , whose objective is 1-strongly convex and 1-smooth. Let  $z^*$  be the minimiser,  $\delta^* = \|z^*\|$ ,  $h(z) = F(z) - F(z^*)$ ,  $M = \text{diam}(Z)$  and  $\delta = \text{PW}(A) > 0$  the pyramidal width [3] of the atom set  $A = \{v_i - w_j\}$  (a property of the  $nm$  atoms, including non-extreme ones, not of  $Z$  alone).

The algorithm keeps simplex weights  $\alpha$  on  $V$ ,  $\beta$  on  $W$ , with  $p = \sum \alpha_i v_i$ ,  $q = \sum \beta_j w_j$ ; the induced product representation  $\lambda_{ij} = \alpha_i \beta_j$  of  $z$  over the atoms  $v_i - w_j$  has support  $\text{supp } \alpha \times \text{supp } \beta$ . With  $g = -\nabla F(z) = q - p$ , the *side scores* are  $s_i^V = \langle g, v_i \rangle$  and  $s_j^W = \langle p - q, w_j \rangle$ ; the implementation maintains them incrementally as differences of cached arrays  $Vp, Vq, Wp, Wq$ .

**Certificate.** For any iterate,  $\text{UB} = \|z\|$  and  $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\| = (\min_i \langle z, v_i \rangle - \max_j \langle z, w_j \rangle) / \text{UB}$  satisfy  $\text{LB} \leq \delta^* \leq \text{UB}$ ; the TA stops when  $\text{UB} - \text{LB} \leq \varepsilon \text{UB}$ . The  $L_2$  soft margin with regularisation parameter  $C$ , the  $\nu$ -SVM and the kernel variants reduce to (1) exactly (augmented coordinates that turn the Gram matrix  $K$  into  $K + I/C$ ; reduced convex hulls; feature-space images), see Appendix B.

**Gaps.**  $g^{\text{FW}} = \max_{u \in Z} \langle g, u \rangle - \langle g, z \rangle$ ,  $g^A = \langle g, z \rangle - \min_{u \in \text{supp } \lambda} \langle g, u \rangle$ ,  $g^{\text{PW}} = g^{\text{FW}} + g^A$ . Because  $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$  is separable and the support is a product set, each gap splits into a  $V$ -side and a  $W$ -side term computable from the score arrays in  $O(n + m)$  (Lemma 4), and a transfer  $v_r - v_u$  on the  $V$  side is a pairwise direction of  $Z$  (Lemma 5).

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**Algorithm 1** One iteration of the guarded block-transfer step (side  $V$ ;  $W$  symmetric)

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1: top- $k$  receivers by  $s^V$ ; worst- $k$  donors in  $\text{supp } \alpha$ ; pair best-with-worst:  $(r_j, u_j)$ ,  $j \leq k'$  ( $k' \leq k$ 
   pairs, limited by the support size); pair 1 is the MDM pair  $(r^*, u^*)$ 
2:  $d_j \leftarrow v_{r_j} - v_{u_j}$ ;  $\text{gap}_j \leftarrow s_{r_j}^V - s_{u_j}^V$ ; discard pairs with  $\text{gap}_j \leq 0$  and receivers that are donors;
    $\gamma_j \leftarrow \min(\text{gap}_j / \|d_j\|^2, c_j)$   $\triangleright c_j$ : donor capacity
3: if pair 1 uncapped ( $\gamma_1 = \text{gap}_1 / \|d_1\|^2$ ) then
4:    $\Delta_B \leftarrow$  gain of exact line search  $t \in [0, 1]$  along  $d = \sum_j \gamma_j d_j$ ;  $\Delta_1 \leftarrow$  gain of the MDM
   step
5:   take the step with the larger gain  $\triangleright$  guard,  $O(k'^2)$ 
6: else
7:   if the away step on  $u^*$  is boundary-clipped then take it  $\triangleright$  drop step:  $\alpha_{u^*} \rightarrow 0$ 
8:   else take the best of {block, MDM, away( $u^*$ ), toward( $r^*$ )}
9:   end if
10: end if

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### 3 Guarded block transfers

Algorithm 1 amortises one  $O(n)$  scan over up to  $k$  transfers. Only the scan is amortised: a new receiver still costs one Gram or kernel column,  $O(nd)$ , and a block touches up to  $2k'$  columns, so Section 5 reports iterations, column evaluations and wall-clock separately. Every candidate gain is available from cached quantities at a cost independent of  $n$ : with  $S = \langle g, d \rangle$  and  $D = \|d\|^2$ , the block gain is  $tS - t^2D/2$  at  $t = \min(S/D, 1)$ , where  $S = \sum_j \gamma_j \text{gap}_j$  and  $D = \gamma^\top \Gamma \gamma$  for the  $k' \times k'$  Gram matrix  $\Gamma$  of the pair directions,  $O(k'^2)$  dot products read off the cached columns; the MDM, away and toward gains are the standard one-dimensional expressions. Throughout, the gain of a step is the decrease  $h(z_t) - h(z_{t+1})$  in the objective. The cache update for the whole block is a single matrix–vector product with the net per-index coefficients; Gram or kernel columns are computed on first use and cached for the run, so support columns cost nothing after their first appearance. Feasibility is immediate: donors lose at most  $t\gamma_j \leq c_j$ , receivers only gain, and the weight sum is conserved.

**Safe screening.** With  $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$  and  $v_{\min} = \arg \min_i \langle z, v_i \rangle$ , a zero-weight point with  $\langle z, v_i \rangle - \langle z, v_{\min} \rangle > r \|v_i - v_{\min}\|$  is outside the optimal support and is removed (symmetrically on  $W$  with the max). Because the optimum’s support survives, the reduced problem has the same optimum and all subsequent lower bounds remain valid.

### 4 Analysis

Full proofs are in Appendix A; we state the results and the ideas.

**Lemma 1** (Block gain). *Let  $b_j = \gamma_j \|d_j\|$  over the  $k'$  retained pairs, all with  $\text{gap}_j > 0$ . Then  $S \geq \sum_j b_j^2$ ,  $D \leq k' S$ , and the realised block gain satisfies  $\Delta_B \geq S/(2k') \geq \Delta_{\text{MDM}}/(2k')$ .*

The proof is the triangle inequality on  $\|d\|$  and Cauchy–Schwarz on  $(\sum b_j)^2$ ; capacity clipping only helps. With the guard, every case-(a) iteration’s gain is  $\max(\Delta_B, \Delta_1) \geq \Delta_1$ : the guarded step never makes less progress than pairwise FW.

**Lemma 2** (Good-step progress). *Work on the side with the larger pairwise gap. On any non-drop iteration,  $\text{gain} \geq \min(g^{\text{PW}^2}/(32M^2), g^{\text{PW}}/8)$ .*

Case (a) uses  $g_{\text{side}}^{\text{PW}} \geq g^{\text{PW}}/2$  and the exact line search bound; case (b) uses  $g_{\text{side}}^{\text{PW}} = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$ , so one of the toward step (always feasible with step  $\leq 1$ ) or the interior away step realises at least a quarter of  $g^{\text{PW}}$ .

**Theorem 1** (Global linear convergence). *Suppose the guarded block-transfer algorithm with away fallback is run for one side per iteration (the side with the larger pairwise gap), using any  $k \geq 1$ , from any vertex pair (initial support  $s_0 \leq 2$ ). Let  $h_t = h(z_t)$ . Then every non-drop iteration satisfies  $h_{t+1} \leq (1 - \rho)h_t$  with*

$$\rho = \frac{1}{16} (\delta/M)^2.$$

Among the first  $T$  iterations, the number of drop steps is at most  $kT_{\text{good}} + s_0$ , where  $T_{\text{good}}$  is the number of non-drop iterations; consequently,  $h_T \leq h_0(1 - \rho)^{(T-s_0)/(k+1)}$ . Furthermore, the two-sided schedule of the implementation (larger-gap side first, second side skipped after a drop) satisfies the same statement with the same  $\rho$  and exponent  $(T - s_0)/(2k + 1)$  (Lemma 7).

The rate follows from Lemma 2 and the geometric strong-convexity bound  $g^{\text{PW}} \geq \delta\sqrt{2h}$  (Lemma 6, from the pyramidal-width definition of [3] and 1-strong convexity), together with  $g^{\text{PW}} \geq g^{\text{FW}} \geq h$  for the linear branch. The drop counting is the away-step argument: each drop removes one support index, non-drop steps add at most  $k$ , so drops are at most  $k$  times the good steps plus  $s_0$ . There is no swap-step bound because case (b) never takes a capacity-clipped pairwise step: it takes an away step, which either contracts or drops. The implementation steps both sides per iteration with the schedule of Lemma 7 (larger-gap side first, second side skipped after a drop); the batteries of §5 were run with the earlier unconditional  $V$ -then- $W$  order, which differs only on drop iterations and agrees with it on the instances of Appendix C.

**Proposition 1** (Near-orthogonal speed-up). *If all  $k'$  pairs are uncapped, the block line search is interior, and  $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$  for  $i \neq j$ , then the block gain is at least  $(\sum_j \Delta_j)/(1 + \eta(k' - 1))$ , where  $\Delta_j$  is pair  $j$ 's individual gain.*

As  $\eta \rightarrow 0$  a single scan realises the sum of all  $k$  pairwise gains; in high dimension distinct support directions are nearly orthogonal, which is the mechanism behind the measured 4–30 $\times$  iteration reductions.

**Lemma 3** (The certified gap is the FW gap). *At any full scan,  $\text{UB} - \text{LB} = g^{\text{FW}}(z)/\text{UB}$ , and  $g^{\text{FW}}(z) \leq 2h + R\sqrt{2h}$  with  $R = \max_{u \in Z} \|u\|$ .*

**Theorem 2** (Screening activation). *Assume  $\delta^* > 0$ . Let  $F_V$  and  $F_W$  be the optimal faces (the vertices that can carry weight at the optimum). Let  $\tau_i = \langle z^*, v_i \rangle - \min_j \langle z^*, v_j \rangle$  be the score margin of  $v_i$  (and symmetrically on  $W$ ), with  $\tau > 0$  the minimal nonzero margin. Finally, let  $L$  be the larger class diameter.*

*A zero-weight point with margin  $\tau_i$  is screened as soon as  $r < \tau_i/(3L)$ . Consequently, once  $h_t < \tau^4 \delta^{*2}/(2592 R^4 L^4)$ , every zero-weight point outside  $F_V \cup F_W$  is removed permanently. Under Theorem 1, this condition holds for all  $t > s_0 + \frac{k+1}{\rho} \log(2592 R^4 L^4 h_0/(\tau^4 \delta^{*2}))$ . Following this iteration, each subsequent scan costs  $O(|F_V| + |F_W| + |\text{supp } \lambda|)$  instead of  $O(n + m)$ .*

In words: since  $r^2 \leq 2R(\text{UB} - \text{LB})$ , a point with margin  $\tau_i$  is screened once the certified gap  $\text{UB} - \text{LB}$  falls below  $\tau_i^2/(18RL^2)$ . We expect the  $\tau$ -dependence to be unavoidable, since a rule that is safe for every instance consistent with a certificate of gap  $\sim \tau$  should not be able to distinguish a point of margin  $\tau$  from a support point; we do not prove this. Each of Lemma 1, Proposition 1, Lemma 3, Lemma 7 and Theorem 2 is verified numerically on random and adversarial ensembles at machine precision (Appendix E).

## 5 Experiments

**Setup.** All solvers are validated against an exact quadratic program (QP; SLSQP over the two simplices) on small instances: distances agree to  $10^{-9}$  or better for every variant. Stopping rule  $\varepsilon = 10^{-3}$  on the certified relative gap throughout unless stated; baselines use matched tolerances on the same objective (Appendix C). Synthetic instances follow the protocol of [2] (two Gaussian clouds,  $n = 5,000$  per class). Real data are the LIBSVM sets a9a, w8a, ijcn1, gisette and covtype (100k subsample), standardised, with official test sets where they exist, five seeds, and a 600 s budget per solver and cell. All real-data runs share one dedicated 32-vCPU Xeon 2.6 GHz machine (122 GB, OpenBLAS, one BLAS thread per solver, 26 cells concurrently), so times are comparable within a battery. Every ETA-versus-baseline pair solves the *same* objective; a differently regularised solver appears only as a labelled reference. “Our SMO” is our own SMO implementation on the same objective; K-ETA is the kernelised ETA.

**Blocks and screening (Appendix C).** On synthetic data block transfers cut iterations 4–30 $\times$  relative to single MDM (plain pairwise FW) but wall-clock only 1.2–2.8 $\times$ , since each block iteration touches more columns (Table 7). On real data (Table 6) blocks cut iterations 6 $\times$  on gisette’s  $L_2$  margin with no wall-clock change, because its 5,000-dimensional columns dominate, and 9 $\times$  on

Table 1: Regime battery on real data: hard margins and kernels, five seeds, 600 s per solver, mean  $\pm$  95% CI. Kernel cells (radial basis function kernel, RBF,  $\gamma = 1/d$ ) use a class-balanced subsample of  $n_k$  training points. ETA and its baseline solve the same objective: LIN, linear hard margin (ETA with screening vs LIBSVM at  $C=10^6$ ; our SMO in brackets); KHM, feature-space hard margin (K-ETA vs LIBSVM at  $C=10^6$ ); KL2, kernel  $L_2$  margin  $K + I/C$ ,  $C=1$  (K-ETA vs kernel SMO). conv = seeds in which ETA reached the certified gap (or certified intersection); oracle =  $O(nd)$  column (ETA) or row (SMO) evaluations, in thousands. On datasets with fixed splits (all but the kernel subsamples) the CIs measure timing noise only.

| cell | dataset | $n_k / d$   | ETA (s)         | baseline (s)                   | conv | dist. ETA / base       | acc. ETA / base | oracle     |
|------|---------|-------------|-----------------|--------------------------------|------|------------------------|-----------------|------------|
| LIN  | gisette | 6000 / 5000 | 56.6 $\pm$ 5.9  | 51.4 $\pm$ 13.1 [405 $\pm$ 62] | 5/5  | 3.6638 / 3.6640        | .981 / .981     | 1.5 [11.1] |
| KHM  | gisette | 6000 / 5000 | 86.2 $\pm$ 14.8 | 104.6 $\pm$ 20.2               | 5/5  | .05215 / .05215        | .979 / .979     | 2.6        |
| KHM  | ijcnn1  | 9853 / 22   | 421 $\pm$ 149   | 1.3 $\pm$ 0.3                  | 4/5  | .00430 / .00430        | .946 / .946     | 2.5        |
| KHM  | a9a     | 10000 / 123 | > 600           | 26.4 $\pm$ 2.9                 | 0/5  | $\leq$ .00009 / .0033  | .690 / .752     | 5.9        |
| KHM  | covtype | 10000 / 54  | > 600           | 144 $\pm$ 7 (maxiter)          | 0/5  | $\leq$ .00053 / .00025 | .704 / .712     | 7.2        |
| KHM  | w8a     | 6479 / 300  | 101 $\pm$ 159   | —                              | 5/5  | hulls intersect        | —               | 1.9        |
| KL2  | gisette | 6000 / 5000 | 131 $\pm$ 34    | 420 $\pm$ 49                   | 5/5  | .06882 / .06882        | .971 / .971     | 3.6 / 15.2 |
| KL2  | w8a     | 6479 / 300  | 11.5 $\pm$ 2.6  | 41.7 $\pm$ 10.0                | 5/5  | .05682 / .05682        | .975 / .975     | 3.6 / 20.2 |
| KL2  | a9a     | 10000 / 123 | 52.8 $\pm$ 71.0 | 57.7 $\pm$ 23.2                | 5/5  | .03019 / .03019        | .795 / .795     | 7.6 / 43.6 |
| KL2  | covtype | 10000 / 54  | 26.0 $\pm$ 4.2  | 17.6 $\pm$ 0.5                 | 5/5  | .02609 / .02609        | .792 / .792     | 9.1 / 56.2 |
| KL2  | ijcnn1  | 9853 / 22   | 9.2 $\pm$ 2.5   | 6.1 $\pm$ 0.7                  | 5/5  | .04790 / .04790        | .959 / .959     | 4.6 / 26.1 |

164 ijcnn1’s kernel cell with a  $2.3\times$  wall-clock gain, because there the  $O(n)$  scan dominates: the block  
165 amortises the scan, not the columns. Screening gives  $1.4\text{--}2.4\times$  at  $\varepsilon = 10^{-5}$  and is neutral at  $10^{-3}$ ,  
166 as Theorem 2 predicts.

167 **Real data, sparse-support regime (Table 1).** TA I first certifies linear separability: gisette is  
168 separable and a9a, w8a, ijcnn1 and covtype are not (hulls intersect, certified in 0.2–42 s), a diagnostic  
169 none of the baselines produces. On gisette ( $d=5,000$ ; 1,233 of 6,000 points in the support) ETA ties  
170 LIBSVM on the linear hard margin, both  $7\times$  ahead of our SMO; K-ETA is  $1.2\times$  faster than LIBSVM  
171 on the RBF hard margin and  $3.2\times$  faster than kernel SMO on the kernel  $L_2$  margin. Distances agree  
172 to five digits, support counts within three, test accuracies exactly. Across all five KL2 cells K-ETA  
173 returns the same distance and accuracy as kernel SMO with 4–6 $\times$  fewer kernel-column evaluations;  
174 in wall-clock it is 3–4 $\times$  faster where columns are expensive (gisette, w8a) and  $1.5\times$  slower where  
175 they are cheap (ijcnn1  $d=22$ , covtype  $d=54$ ), because the  $O(n)$  NumPy scan per iteration then  
176 dominates. The reference hinge-loss SVC ( $C=1$ , a different objective) takes 1–84 s with accuracies  
177 within one point (Appendix C).

178 The KHM column shows where the geometry turns against the method. On ijcnn1 both solvers  
179 return the identical solution (distance 0.00430, 819 support vectors, accuracy 0.946), but K-ETA  
180 needs  $1.5\text{--}2.5 \times 10^5$  iterations and 420 s against LIBSVM’s 1.3 s: the feature-space hulls are 0.004  
181 apart in a space of diameter  $\approx \sqrt{2}$ , so  $\delta/M$  in Theorem 1 is of order  $10^{-3}$ , and LIBSVM’s second-  
182 order working-set selection with heuristic shrinking is far better suited to that conditioning. On a9a  
183 and covtype the hulls are within  $10^{-4}$  and  $5 \times 10^{-4}$  of touching: K-ETA times out at gap 3–10, while  
184 LIBSVM’s  $C=10^6$  surrogate stops at its Karush–Kuhn–Tucker (KKT) tolerance at a point 6–30 $\times$   
185 farther out than K-ETA’s upper bound (or hits its iteration cap on covtype), so neither solver certifies  
186 anything. K-ETA’s closer iterate has the lower test accuracy on a9a (0.690 vs 0.752) because the  
187 classifier is the bisector of  $p - q$ , whose direction is ill-determined when  $\|p - q\| \rightarrow 0$ ; the hard-  
188 margin kernel model is the wrong one there anyway (KL2 accuracies 0.79 vs 0.70–0.75). On w8a  
189 the feature-space hulls intersect (duplicate points with opposite labels); K-ETA certifies this in 101 s  
190 and no hard margin exists.

191 **Real data,  $L_2$  soft margins (Table 2, appendix).** The same boundary appears within one problem  
192 class. Gisette is separable, so its  $L_2$  margin has a sparse support (1,233 of 6,000 points) and ETA  
193 converges to the certified gap in 30–45 s at every  $C$ ,  $7\times$  faster than our SMO on the same objective.  
194 LIBLINEAR run to the same primal accuracy (tolerance  $10^{-6}$ ; at  $C=1$ : 0.14900 vs ETA’s 0.14893,  
195 inside ETA’s certified  $8 \times 10^{-4}$ ) takes 8 min to 1.5 h, but that is the tolerance needed to match  
196 ETA’s primal, not a property of the problem: at its default tolerance ( $10^{-4}$ ) LIBLINEAR takes  
197 5–66 s, comparable to ETA, but stops 5–24% above the certified optimum with test accuracy 0.973–  
198 0.978 against 0.981 (Table 4). We therefore claim no wall-clock advantage over LIBLINEAR on

gisette; what ETA adds is the certificate, which reveals that the fast answer is not the optimum. Even at  $10^{-6}$ , at  $C=10$  LIBLINEAR stops 5% above the optimum (0.1566 vs 0.1490), since on separable data the primal is nearly flat at large  $C$ . By the utility metric of time to 1% relative primal suboptimality (Table 5), ETA’s iterate qualifies after 34–44 s, a few seconds before it certifies  $10^{-3}$ , whereas LIBLINEAR’s default-tolerance answers never qualify. On the four overlapping datasets both geometric solvers hit the 600 s cap on every cell while LIBLINEAR finishes in 0.3–39 s. At  $C=0.1$  ETA’s iterate already has LIBLINEAR’s test accuracy on a9a, ijcnn1 and w8a (0.849, 0.918, 0.987) when the cap is reached with the certificate still at 0.01–0.6: the iterate converges long before the certificate closes. On the uncapped a9a and w8a runs the iterate is within 1% of the optimum after 811 and 2,084 s, against 3,030 and 5,386 s to certify  $10^{-3}$  and LIBLINEAR’s 4 and 23 s (Table 5). At  $C \geq 1$  the augmented hulls  $(K + I/C)$  approach each other, the certified gap at the cap exceeds one and the iterate is not yet usable; on covtype (100,000 points) it is not usable at any  $C$ . Uncapped, ETA does converge on the overlapping sets: on a9a ( $C=0.1$ , ten runs) in  $2,908 \pm 138$  s and 95,201 iterations, with primal objective 687.07, identical to LIBLINEAR’s to five digits, and on w8a in 5,153 s. Figure 1 (appendix) shows these traces: the linear-rate signature of Theorem 1, but with 60–65% of the points in the support and near-uniform weights: the active atoms form a near-regular simplex on  $s \approx 0.6n$  vertices, whose pyramidal width is  $2/\sqrt{s}$  times its side [3], so  $\rho \sim 1/s$ , the iteration count grows linearly in  $n$  and, at  $O(n)$  per iteration, the cost is  $O(n^2)$ -type. Centroid initialisation and larger blocks did not help.

## 6 Related work

Distance duality and the TA [1, 2]; MDM [4] and its linear rate [24]; Gilbert’s algorithm [5] and Wolfe’s method [25]; FW for polytope distance and SVMs [20, 21, 22, 23]; the nearest-point and reduced-hull view of soft margins [6, 8]; away steps [7] and the pyramidal-width analysis [3], whose pairwise step is our MDM pair, with the facial distance of [26] as a computable alternative to  $\delta$ . Swap-step-free pairwise variants [12, 13] and block-coordinate FW [9] are the closest algorithmic relatives; gap-safe screening [14, 16, 17] and active-set identification for FW [18, 19] the closest to Theorem 2, with [10, 11, 15] as its SVM ancestors. Appendix D details the differences.

## 7 Conclusion

Guarded block transfers give the Triangle Algorithm a linear rate with explicit constants and no combinatorial swap-step bounds, together with a screening rule that is both safe and provably effective. The cost is a factor  $(k + 1)$  in the exponent, recovered in the near-orthogonal regime where a block realises close to the sum of its  $k$  pairwise gains. The experiments locate the boundary the analysis predicts: sparse support and well-separated hulls, where the solver matches or beats LIBSVM and kernel SMO and supplies a certificate, against dense support or nearly touching hulls, where  $\delta/M$  is small and primal coordinate descent or second-order SMO is faster by one to two orders of magnitude. Which side a run is on can be read from its certified gap and support size. Limitations: screening pays only when the certificate closes; the implementation is NumPy-level (a compiled inner loop would move the practical boundary, not the  $O(n^2)$  asymptotics) without sparse support; kernel experiments use 10k-point subsamples; the reduced-hull ( $\nu$ -SVM) variant is validated but not analysed; the constants in Theorem 1 are loose; and  $k$  is fixed, although Table 6 shows diminishing returns beyond  $k=4$  on expensive-column data, so an adaptive  $k$  is future work.

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## 296 A Proofs

297 **Lemma 4** (Separability). *With the product representation,  $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$  where  $g_V^{\text{PW}} =$   
298  $\max_i s_i^V - \min_{i \in \text{supp } \alpha} s_i^V \geq 0$  and  $g_W^{\text{PW}}$  likewise; the same holds for  $g^{\text{FW}}$  and  $g^{\text{A}}$ .*

299 *Proof.*  $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$  is separable and the support is a product set, so the max over  
300 atoms and the min over the support separate; each side term is nonnegative since the support max  
301 dominates the support min and the global max dominates the support max.  $\square$

302 **Lemma 5** (Side transfers are  $Z$ -pairwise directions). *For  $\alpha_u > 0$  and any  $b \in \text{supp } \beta$ ,  $v_r - v_u =$   
303  $(v_r - w_b) - (v_u - w_b)$  is a difference of an atom of  $Z$  and an atom in the support, and the transfer  
304  $\gamma \leq \alpha_u$  realises  $z \leftarrow z + \gamma(v_r - v_u)$  inside  $Z$ .*

305 *Proof of Lemma 1.* Since  $\gamma_j \leq \text{gap}_j / \|d_j\|^2$ , we have  $\gamma_j \text{gap}_j \geq \gamma_j^2 \|d_j\|^2 = b_j^2$ . Consequently,  
306  $S = \sum_j \gamma_j \text{gap}_j \geq \sum_j b_j^2$ . By the triangle inequality,  $\|d\| \leq \sum_j \gamma_j \|d_j\| = \sum_j b_j$ , and by Cauchy-  
307 Schwarz,  $(\sum_j b_j)^2 \leq k' \sum_j b_j^2 \leq k' S$ ; hence  $D \leq k' S$ . Exact line search on the 1-smooth  $F$   
308 gives  $\Delta_B = S^2 / (2D) \geq S / (2k')$  in the interior case and  $\Delta_B = S - D/2 \geq S/2$  at the boundary  
309  $t = 1$ , where  $D \leq S$ . Finally,  $S \geq \gamma_1 \text{gap}_1 \geq \Delta_{\text{MDM}}$ , since all terms are nonnegative and  
310  $\Delta_{\text{MDM}} = \gamma_1 \text{gap}_1 - \gamma_1^2 \|d_1\|^2 / 2$ .  $\square$

311 *Proof of Lemma 2.* Let  $g_S = \max(g_V^{\text{PW}}, g_W^{\text{PW}}) \geq g^{\text{PW}} / 2$ . Every direction below is a difference of  
312 two points of one class hull, so its norm is at most  $\text{diam}(\text{conv } V) \leq M$ . For a feasible direction with  
313 directional derivative  $\sigma > 0$ , squared norm  $D$  and maximal step  $\bar{t}$ , exact line search on a 1-smooth  
314 function gives gain  $\geq \min(\sigma^2 / (2D), \bar{t}\sigma / 2)$  (interior optimum, or the clipped step evaluated with  
315  $\bar{t}D \leq \sigma$ ). In case (a), the guard's gain is at least the MDM gain, whose derivative is  $g_S$  and whose  
316 interior optimum is attained, yielding gain  $\geq g_S^2 / (2M^2) \geq g^{\text{PW}^2} / (8M^2)$ . In case (b), non-drop,  
317  $g_S = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$ , so one term is at least  $g_S / 2 \geq g^{\text{PW}} / 4$ . If it is the FW term, the toward  
318 step (maximal step 1) gives gain  $\geq \min(g_{\text{side}}^{\text{FW}^2} / (2M^2), g_{\text{side}}^{\text{FW}} / 2) \geq \min(g^{\text{PW}^2} / (32M^2), g^{\text{PW}} / 8)$ ;  
319 if it is the away term, the away line search is interior (non-drop) and gain  $\geq g_{\text{side}}^{\text{A}^2} / (2M^2) \geq$   
320  $g^{\text{PW}^2} / (32M^2)$ . The algorithm takes the best available candidate.  $\square$

321 **Lemma 6** (Geometric strong convexity; [3, Thm. 6] with  $\mu = 1$ ). *For every  $z \in Z$  with a represen-*  
322 *tation whose support contains an away atom,  $g^{\text{PW}}(z) \geq \delta \sqrt{2h(z)}$  and  $g^{\text{PW}}(z) \geq g^{\text{FW}}(z) \geq h(z)$ .*

323 *Proof.* This is Theorem 6 of [3] for a 1-strongly convex objective; we include the short argument.  
324 By the definition of the pyramidal width [3, Def. 3], for  $e = (z^* - z) / \|z^* - z\|$  the FW atom  $s$  and  
325 the away atom  $v$  satisfy  $\langle g, s - v \rangle \geq \delta \langle g, e \rangle$ . By 1-strong convexity,  $\langle g, z^* - z \rangle \geq h + \frac{1}{2} \|z^* - z\|^2$ ,  
326 so with  $\varrho = \|z^* - z\|$ ,  $g^{\text{PW}} \geq \delta(h/\varrho + \varrho/2) \geq \delta \sqrt{2h}$  by AM-GM. Finally  $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}} \geq$   
327  $g^{\text{FW}} = \max_u \langle g, u - z \rangle \geq \langle g, z^* - z \rangle \geq h$  by convexity.  $\square$

328 *Proof of Theorem 1.* By Lemmas 2 and 6 a non-drop iteration gains at least  
329  $\min(2\delta^2 h / (32M^2), h/8) = (\delta/M)^2 h / 16$ , using  $\delta \leq M$ ; hence  $h_{t+1} \leq (1 - \rho)h_t$ . Drop  
330 counting: non-drop iterations add at most  $k$  support indices, drop iterations add none and remove at  
331 least one, and the support size stays  $\geq 2$  from  $s_0$ ; so the number of removals, hence of drops, is at  
332 most  $kT_{\text{good}} + s_0$ , giving  $T_{\text{good}} \geq (T - s_0)/(k + 1)$ . All steps, drops included, are descent steps,  
333 so  $h$  is monotone.  $\square$

334 **Lemma 7** (Two-sided schedule). *Let every iteration compute both side gaps, take the guarded step*  
335 *of Algorithm 1 on the side with the larger pairwise gap first, and take the other side's guarded*  
336 *step only if the first step was not a drop; an iteration counts as a drop iff its first step is a drop.*  
337 *Then every non-drop iteration gains at least  $\min(g^{\text{PW}^2} / (32M^2), g^{\text{PW}} / 8)$  and the number of drops*  
338 *is at most  $2kT_{\text{good}} + s_0$ ; hence Theorem 1 holds for this schedule with the same  $\rho$  and exponent*  
339  *$(T - s_0)/(2k + 1)$ .*

340 *Proof.* On a non-drop iteration the first step is exactly the step analysed in Lemma 2, so it gains at  
341 least the stated amount; the second step is an exact line search along a feasible descent direction (or  
342 is skipped), so it cannot increase  $h$ . Drop counting: a drop iteration takes a single away step that

removes one support index and adds none, because the second side is skipped; a non-drop iteration takes two guarded steps and adds at most  $k$  indices each. Removals, hence drops, therefore number at most  $2kT_{\text{good}} + s_0$ .  $\square$

*Remark 1.* The skip is what makes the count work. If the second side always steps, an iteration whose analysed side drops can add up to  $k$  indices on the other side, and the number of such iterations is then bounded only by  $\frac{k}{k+1}T$ , which gives no rate for  $k \geq 1$ .

*Proof of Proposition 1.*  $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 \leq (1 + \eta(k' - 1))S$  by Cauchy–Schwarz on the cross terms; then  $\Delta_B = S^2/(2D) \geq S/(2(1 + \eta(k' - 1)))$ , and for uncapped pairs  $S = \sum_j 2\Delta_j$ .  $\square$

*Proof of Lemma 3.*  $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\|$  by definition of the supporting hyperplane bound and  $\text{UB} = \|z\|$ ; subtracting,  $\text{UB} - \text{LB} = (\langle z, z \rangle - \min_u \langle z, u \rangle) / \|z\| = g^{\text{FW}} / \text{UB}$ . At the optimum  $0 = \langle z^*, z^* \rangle - \min_u \langle z^*, u \rangle$ ; subtracting from  $g^{\text{FW}}(z)$ ,  $g^{\text{FW}}(z) = (\|z\|^2 - \|z^*\|^2) + (\min_u \langle z^*, u \rangle - \min_u \langle z, u \rangle) \leq 2h + \max_u \langle z^* - z, u \rangle \leq 2h + R\|z - z^*\| \leq 2h + R\sqrt{2h}$  by 1-strong convexity.  $\square$

*Proof of Theorem 2.* Strong convexity gives  $\|z - z^*\| \leq r$  for the screening radius  $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ . Let  $m^* = \min_j \langle z^*, v_j \rangle$  and pick  $v' \in F_V$ ; since  $v_{\min}$  minimises  $\langle z, \cdot \rangle$ ,  $\langle z^*, v_{\min} \rangle - m^* = \langle z, v_{\min} - v' \rangle + \langle z^* - z, v_{\min} - v' \rangle \leq rL$ . Hence  $\langle z, v_i - v_{\min} \rangle = \langle z^*, v_i - v_{\min} \rangle + \langle z - z^*, v_i - v_{\min} \rangle \geq \tau_i - rL - r\|v_i - v_{\min}\|$ , and the rule fires when this exceeds  $r\|v_i - v_{\min}\|$ , for which  $\tau_i > 3rL$  suffices since  $\|v_i - v_{\min}\| \leq L$ . For the activation time,  $r^2 = (\text{UB} - \text{LB})(\text{UB} + \text{LB}) \leq 2R(\text{UB} - \text{LB}) = 2Rg^{\text{FW}}/\text{UB} \leq (2R/\delta^*)(2h + R\sqrt{2h}) \leq (4R^2/\delta^*)\sqrt{2h}$  for  $h \leq R^2/2$  (Lemma 3); requiring  $r^2 \leq \tau^2/(9L^2)$  yields the stated condition on  $h_t$ , and Theorem 1 with  $\log(1/(1-\rho)) \geq \rho$  converts it into the iteration bound. Permanence follows from the safety of the rule.  $\square$

## B Exact reductions

**$L_2$  soft margin.** The problem  $\min \frac{1}{2}\|w\|^2 + \frac{C}{2} \sum_i \xi_i^2$  is the hard-margin problem on the augmented points  $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$ , whose Gram matrix is  $K + I/C$  ( $K$  the Gram matrix of the original points) and whose augmented hulls never intersect; let  $\delta_C$  denote their distance. The augmentation is never materialised: the extra coordinates of an iterate are exactly its convex weights, so every augmented inner product is a cached original-space product plus a sparse diagonal correction,  $\langle \tilde{p}, \tilde{v}_i \rangle = \langle p, v_i \rangle + \alpha_i/C$  and  $\|\tilde{p}\|^2 = \|p\|^2 + \|\alpha\|^2/C$ . The separator is recovered as  $w = 2(p - q)/\delta_C^2$ ,  $b = (\|\tilde{q}\|^2 - \|\tilde{p}\|^2)/\delta_C^2$ , and the primal value is  $P = 2/\delta_C^2$ . Validation: the solver agrees with an exact QP on the explicitly augmented problem to  $10^{-10}$  across random instances and values of  $C$ ; the regularisation path  $\delta_C$  is monotone in  $C$ ; on a9a ( $C=0.1$ ) its certified primal 687.07 equals LIBLINEAR’s to five digits.

**$\nu$ -SVM.** The  $L_1$  (hinge) soft margin is the nearest-point problem between *reduced* convex hulls  $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$ ,  $\mu = 2/(\nu\ell)$  with  $\ell = n + m$  the number of training points [8]. Two changes: the extreme-point oracle becomes a capped top- $k$  blend (weight  $\mu$  on the  $\lfloor 1/\mu \rfloor$  best points in the search direction, via a partial sort, still  $O(n)$ ), and the primary step is the capped MDM transfer with blended toward steps as fallback; caching and the reduced support-function lower bound carry over unchanged. Validation: agreement with the exact box-constrained QP to  $10^{-9}$ – $10^{-14}$  and with LIBSVM’s NuSVC to  $\sim 10^{-10}$  through the mapping  $\delta^* = \|w_{\text{svc}}\|/\varsigma$ ,  $\varsigma$  the per-class sum of the rescaled dual coefficients. The reduced hull is a polytope, so (1) still applies, but its vertices are blends and the capped transfer is not the pairwise step analysed in §4; the analysis of this variant is left open.

**Kernels.** The algorithm touches data only through inner products, so kernelisation changes plumbing, not algorithm: Gram columns become kernel columns  $\kappa(V, x_i)$ , squared norms become kernel diagonals, iterates exist purely as convex weights (the exact cache refresh is reconstructed from the support’s cached kernel columns), and the classifier is the kernel expansion of the witness-pair bisector. A ridge  $K + I/C$  gives the kernel  $L_2$  margin by the same diagonal trick. Validation: the

linear kernel reproduces the Euclidean solver to  $10^{-12}$ ; RBF distances match the exact kernel QP to  $10^{-9}$  and LIBSVM’s hard-margin solution, via  $\|w_{\mathcal{H}}\|^2 = (\alpha \circ y)^\top K(\alpha \circ y)$  for LIBSVM’s dual coefficients  $\alpha$  and labels  $y$ , to  $10^{-5}$ ; kernel SMO matches the QP to  $10^{-8}$ .

## C Experimental details

**Solver settings and tolerance equivalence.** ETA: block transfers with  $k = 16$ , guarded, away fallback, pairwise anti-zig-zag, stop at certified relative gap  $10^{-3}$ , iteration cap  $2 \times 10^6$ , 600 s cap. The batteries were run with the unconditional  $V$ -then- $W$  order; the schedule of Lemma 7 (larger gap first, skip after a drop) differs from it only on drop iterations, and Table 3 shows the two agree on the synthetic instances of this appendix; screening (every 25 iterations) only in the LIN cell, where it never activated at  $10^{-3}$ , as Theorem 2 predicts. Our SMO: maximal-violating-pair selection, KKT gap  $10^{-3}$ , same caps; hard margin as  $C = 10^{12}$ ; kernel SMO on  $K+I/C$ . LIBLINEAR (scikit-learn 1.8 LinearSVC): squared hinge with parameter  $C/2$ , so that its objective equals  $\frac{1}{2}\|w\|^2 + \frac{C}{2} \sum \xi_i^2$ ; tolerance  $10^{-6}$ , chosen so that its final primal matches ETA’s certified value on every cell where ETA converges (a9a 687.07 vs 687.07; gisette 0.14900 vs 0.14893, inside ETA’s certified  $8 \times 10^{-4}$ ; ijcn1 550.6 vs 550.5 at ETA gap 0.01); this tolerance is also why LIBLINEAR needs 18,000 iterations and over an hour on gisette at  $C=1$ , and 35,000 iterations and 1.5 h at  $C=10$ , where it nevertheless stops 5% above the certified optimum (0.1566 vs 0.1490). The gisette LIBLINEAR times in Table 2 are therefore the cost of matching ETA’s primal; Table 4 reports the same configuration at scikit-learn’s default tolerance  $10^{-4}$  (five seeds, run separately without concurrent load): 5–66 s, but primals 5–24% above the certified optimum and lower test accuracy. Intercept scaling 100; iteration cap  $2 \times 10^5$ . LIBSVM (scikit-learn SVC): tolerance  $10^{-3}$ , 1000 MB cache, iteration cap  $2 \times 10^6$  (reported as *maxiter* when hit); hard margins as  $C = 10^6$ ; RBF with  $\gamma = 1/d$ . Oracle counts: ETA counts Gram or kernel columns evaluated against a full class ( $O(nd)$  or  $O(n_k d)$  each); SMO counts kernel rows, the same unit. Columns are cached without eviction and the block Gram matrix  $\Gamma$  is assembled from four cached entries per pair, so peak cache memory is the column count times  $n_k$  doubles: 0.61 GB for the largest cell (a9a KL2, 7,600 columns) and 0.36 GB for ijcn1 KL2.

**Data and seeds.** a9a, w8a, ijcn1 and gisette use the LIBSVM training and test files; covtype (binary, scaled) is subsampled to 100,000 points and split 75/25 by seed. Features are standardised on the training set. Kernel cells use a class-balanced subsample of at most 5,000 points per class (w8a’s minority class has 1,479 points, hence  $n_k = 6,479$ ; gisette uses all 6,000). Seeds vary the covtype split, the kernel subsamples and the timing environment; on datasets with fixed splits the deterministic solvers return identical solutions across seeds, which is why several accuracy CIs are zero. Confidence intervals are  $t$ -based 95% intervals over seeds.

ETA on real data (L2 soft margin,  $C = 0.1$ ): certified gap to the  $10^{-3}$  stopping rule

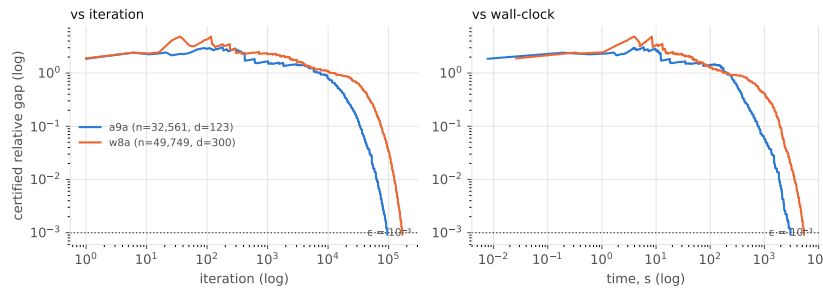


Figure 1: Certified relative gap of ETA on a9a and w8a ( $L_2$ ,  $C = 0.1$ , uncapped), versus iteration and wall-clock: linear-rate convergence in the dense-support regime, at  $O(n^2)$ -type cost.

**Hardware and concurrency.** One dedicated 32-vCPU Intel Xeon 2.6 GHz Lightning Studio, 122 GB RAM, Linux, Python 3.12, NumPy 2.4 with OpenBLAS 0.3.31, OMP\_NUM\_THREADS=1; the  $L_2$  battery ran 14 cells and the regime battery 12 cells concurrently (the gisette, covtype and w8a  $C=10$  cells of Table 2 were completed in later passes with 11–14 workers and no other load; the

runs of Tables 4 and 6 were made separately with no concurrent load, which is why their gisette  $k=16$  time is below Table 2’s). The uncapped a9a/w8a runs in Figure 1 were made earlier on the same machine under a different load (LIBLINEAR took 12 and 48 s there against 4 and 23 s in the capped battery), so ratios across the two runs carry a factor of roughly two of uncertainty; ratios within a battery do not.

Table 2:  $L_2$  soft-margin battery: five seeds, 600 s per solver for ETA and our SMO (our own implementation; LIBSVM does not solve this objective), LIBLINEAR at tolerance  $10^{-6}$  (Appendix C) and unbounded; mean  $\pm$  95% CI, which on the fixed-split datasets (all but covtype) measures timing noise only. “cap” = timed out in every seed; ETA’s certified relative gap is reported at its stopping point (mean [min, max]); accuracies are those of the returned iterate.

| dataset ( $n / d$ )     | $C$ | ETA (s)        | SMO (s)       | LIBLINEAR (s)   | ETA gap at stop      | acc. ETA / SMO / LIBLINEAR |
|-------------------------|-----|----------------|---------------|-----------------|----------------------|----------------------------|
| gisette (6,000 / 5,000) | 0.1 | 35.3 $\pm$ 6.8 | 243 $\pm$ 53  | 500 $\pm$ 41    | $8 \times 10^{-4}$   | .981 / .981 / .981         |
|                         | 1   | 43.3 $\pm$ 5.3 | 295 $\pm$ 6   | 3916 $\pm$ 115  | $8 \times 10^{-4}$   | .981 / .981 / .981         |
|                         | 10  | 33.2 $\pm$ 4.6 | 236 $\pm$ 36  | 5396 $\pm$ 197  | $10 \times 10^{-4}$  | .981 / .981 / .977         |
| a9a (32,561 / 123)      | 0.1 | cap            | cap           | 3.7 $\pm$ 0.8   | 0.15 [0.10, 0.23]    | .849 / .845 / .849         |
|                         | 1   | cap            | cap           | 6.8 $\pm$ 1.7   | 1.26 [1.25, 1.28]    | .810 / .780 / .849         |
|                         | 10  | cap            | cap           | 2.3 $\pm$ 0.5   | 1.83 [1.80, 1.86]    | .635 / .763 / .849         |
| ijcnn1 (49,990 / 22)    | 0.1 | cap            | 478 $\pm$ 181 | 0.3 $\pm$ 0.1   | 0.010 [0.005, 0.016] | .918 / .918 / .918         |
|                         | 1   | cap            | cap           | 0.6 $\pm$ 0.2   | 0.68 [0.61, 0.73]    | .901 / .917 / .918         |
|                         | 10  | cap            | cap           | 0.4 $\pm$ 0.1   | 1.34 [1.27, 1.40]    | .656 / .852 / .918         |
| w8a (49,749 / 300)      | 0.1 | cap            | cap           | 22.7 $\pm$ 4.3  | 0.57 [0.50, 0.66]    | .987 / .985 / .987         |
|                         | 1   | cap            | cap           | 20.6 $\pm$ 6.6  | 1.17 [1.14, 1.21]    | .923 / .979 / .987         |
|                         | 10  | cap            | cap           | 13.2 $\pm$ 1.1  | 2.02 [1.94, 2.14]    | .720 / .971 / .987         |
| covtype (100,000 / 54)  | 0.1 | cap            | cap           | 19.4 $\pm$ 15.7 | 1.04 [1.02, 1.05]    | .568 / .660 / .756         |
|                         | 1   | cap            | cap           | 6.1 $\pm$ 3.6   | 1.42 [1.23, 1.73]    | .508 / .541 / .756         |
|                         | 10  | cap            | cap           | 5.0 $\pm$ 1.8   | 2.01 [1.20, 2.37]    | .517 / .514 / .756         |

**Reference solver in the KL2 cells.** The hinge-loss SVC(rbf,  $C=1$ ), a different objective, took 8.6  $\pm$  2.9 (a9a), 3.4  $\pm$  0.3 (covtype), 83.5  $\pm$  13.1 (gisette), 1.1  $\pm$  0.0 (ijcnn1) and 4.9  $\pm$  1.2 s (w8a), with test accuracies 0.789, 0.783, 0.977, 0.950 and 0.975, within one point of the  $L_2$  solutions and on either side of them.

**Step ablation, blocks and shrinking (synthetic).** Iteration counts at  $\varepsilon = 10^{-6}$  (single instances):  $d=30$ ,  $n=200$ : toward 158,941, midpoint 148,996, away 106, pairwise 181;  $d=100$ ,  $n=1,000$ : toward and midpoint fail within 300,000, away 15,886, pairwise 3,921. At the working tolerance  $10^{-3}$  the pairwise/away gain over the midpoint heuristic is only  $\sim 11\%$ , which is why the heuristic sufficed in [2]. Block transfers cut iterations 4–30 $\times$  relative to single MDM (e.g. 43,486  $\rightarrow$  1,551 on a dense-support  $L_2$  instance at  $d=1,000$ ) and are the only configuration that reaches the certified tolerance on MNIST odd-vs-even. Figure 2 shows the screening speed-ups and Figure 3 the consolidated comparison: hard margins 0.24 vs LIBSVM’s 0.39 s at  $d=1,000$  and 8.6 vs 20.5 s at  $d=10,000$ ; 0.21 vs 0.42 s at  $\varepsilon=10^{-5}$  (where the paper-style configuration fails to converge); RBF hard margin 0.76 vs 1.28 s;  $\nu$ -SVM 18.5 vs 21.9 s; synthetic  $L_2$  ( $d=1,000$ ,  $C=1$ ) 8.9 vs LIBLINEAR’s 13.8 s; MNIST odd-vs-even 82.8 vs LIBLINEAR’s 55.0 s. Distances agree within tolerance on every row.

Table 3: Two-sided schedules on the synthetic instances of this appendix ( $n=5,000$  per class,  $k=16$ ): the unconditional  $V$ -then- $W$  order used for the batteries versus the schedule of Lemma 7 (larger-gap side first, second side skipped after a drop). Drop steps and skipped second-side steps are counted over the run.

| instance                                      | schedule      | time (s) | iterations | drops | skips | distance  |
|-----------------------------------------------|---------------|----------|------------|-------|-------|-----------|
| hard margin, $d=1000$ , $\varepsilon=10^{-5}$ | unconditional | 0.1      | 16         | 0     | 0     | 55.339365 |
| hard margin, $d=1000$ , $\varepsilon=10^{-5}$ | Lemma 7       | 0.2      | 16         | 0     | 0     | 55.339365 |
| $L_2$ overlap, $d=100$ , $C=1$                | unconditional | 7.9      | 11,216     | 638   | 0     | 0.078400  |
| $L_2$ overlap, $d=100$ , $C=1$                | Lemma 7       | 8.1      | 10,701     | 599   | 316   | 0.078400  |
| $L_2$ overlap, $d=1000$ , $C=1$               | unconditional | 4.3      | 2,951      | 172   | 0     | 0.578143  |
| $L_2$ overlap, $d=1000$ , $C=1$               | Lemma 7       | 4.3      | 3,001      | 180   | 121   | 0.578143  |

Table 6 was produced by the current code, whose default is the schedule of Lemma 7; its distances and accuracies coincide with those of Tables 1 and 2, which used the unconditional order, so the two schedules agree on real data as on the synthetic instances of Table 3. Away fallbacks that drop an

Table 4: LIBLINEAR on gisette at its default tolerance ( $10^{-4}$ ) versus the matched tolerance ( $10^{-6}$ ) of Table 2; five seeds. ETA’s certified primal is 0.14835 / 0.14893 / 0.14899 at  $C = 0.1 / 1 / 10$ , reached in 30–45 s with test accuracy 0.981.

| $C$ | tol       | time (s)       | iterations | primal (vs certified) | accuracy |
|-----|-----------|----------------|------------|-----------------------|----------|
| 0.1 | $10^{-4}$ | $66 \pm 10$    | 363        | 0.15648 (+5.5%)       | .978     |
| 0.1 | $10^{-6}$ | $500 \pm 41$   | 2,434      | 0.14835 (+0.0%)       | .981     |
| 1   | $10^{-4}$ | $5.5 \pm 0.8$  | 10         | 0.18427 (+24%)        | .974     |
| 1   | $10^{-6}$ | $3916 \pm 115$ | 18,063     | 0.14900 (+0.05%)      | .981     |
| 10  | $10^{-4}$ | $4.8 \pm 0.1$  | 8          | 0.18487 (+24%)        | .973     |
| 10  | $10^{-6}$ | $5396 \pm 197$ | 35,411     | 0.15663 (+5.1%)       | .977     |

Table 5: Time to 1% relative primal suboptimality, seed-0 traces. For ETA the certified dual value  $2/\text{UB}_t^2$  increases to the optimum, so  $s_t = 1 - (2/\text{UB}_t^2)/P^*$  with  $P^*$  the best known primal (the lower of LIBLINEAR’s  $10^{-6}$  value and ETA’s certified value); “certified 1%” is the first time the certificate itself proves  $1 - \text{LB}_t^2/\text{UB}_t^2 \leq 1\%$ . ETA’s a9a and w8a times are from the uncapped runs of Figure 1. For LIBLINEAR the entry is the time of the coarsest tolerance whose primal is within 1% of  $P^*$ ; where only  $10^{-6}$  has been run it is an upper bound. On the remaining cells ETA is not within 1% at the 600 s cap ( $s = 30$ –99%).

| dataset | $C$ | $P^*$   | ETA 1%  | ETA 0.1% | cert. 1% | cert. $10^{-3}$        | LIBLINEAR 1%                   |
|---------|-----|---------|---------|----------|----------|------------------------|--------------------------------|
| gisette | 0.1 | 0.14835 | 40 s    | 42 s     | 43 s     | 43 s                   | $\leq 500$ s (tol $10^{-6}$ )  |
| gisette | 1   | 0.14893 | 44 s    | 46 s     | 47 s     | 48 s                   | $\leq 3916$ s (tol $10^{-6}$ ) |
| gisette | 10  | 0.14899 | 34 s    | 36 s     | 37 s     | 37 s                   | none at $10^{-4}$ or $10^{-6}$ |
| a9a     | 0.1 | 687.07  | 811 s   | 1,413 s  | 2,121 s  | 3,030 s                | 4 s                            |
| w8a     | 0.1 | 108.80  | 2,084 s | 2,960 s  | 4,162 s  | 5,386 s                | 23 s                           |
| ijcnn1  | 0.1 | 550.61  | 301 s   | 472 s    | > 600 s  | > 600 s ( $s=0.05\%$ ) | 0.3 s                          |

index occur on 1–36% of block iterations and never with single transfers, and their share grows with  $k$ : larger blocks add more indices per iteration, and the drops remove them.

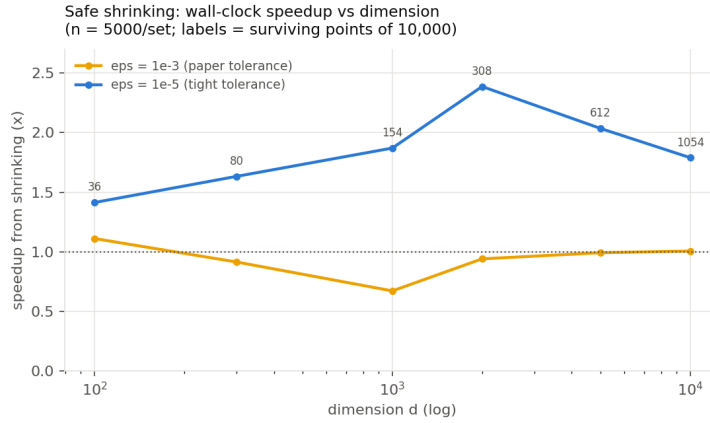


Figure 2: Safe screening: wall-clock speed-up vs dimension at two tolerances on the separated synthetic protocol; labels are surviving points of 10,000.

## 453 D Extended related work

454 **Swap steps.** Blended pairwise conditional gradients [12] also remove pairwise FW’s swap-step  
455 bound with a guard, there between the pairwise step and a FW step, and [13] avoid bad steps with  
456 a short-step chain. Our guard compares an aggregated block of  $k$  transfers with the single MDM  
457 step and falls back to an away step, so the drop-step count of away-step FW applies unchanged.  
458 Block-coordinate FW [9] blocks over *coordinates* of a product domain rather than aggregating pairs  
459 on one polytope.

Table 6: Block-size ablation on real data (ETA only, three seeds, run separately without concurrent load): the gisette  $L_2$  margin ( $C=1$ ,  $d=5,000$ ) and the ijcnl kernel  $L_2$  cell of Table 1 ( $d=22$ ,  $n_k=9,853$ ).  $k=1$  is plain pairwise FW (single MDM transfers, no block, no guard); drops = away fallbacks that removed a support index. Every cell converged to the certified gap with identical distance and accuracy per dataset.

| instance                   | $k$ | time (s)       | iterations | columns | drops |
|----------------------------|-----|----------------|------------|---------|-------|
| gisette $L_2$ , $C=1$      | 1   | $26.3 \pm 5.4$ | 4,946      | 1,300   | 0     |
| gisette $L_2$ , $C=1$      | 4   | $25.3 \pm 5.1$ | 1,071      | 1,349   | 80    |
| gisette $L_2$ , $C=1$      | 16  | $27.8 \pm 4.4$ | 831        | 1,480   | 301   |
| ijcnl kernel $L_2$ , $C=1$ | 1   | $13.6 \pm 0.4$ | 12,796     | 4,520   | 0     |
| ijcnl kernel $L_2$ , $C=1$ | 4   | $6.6 \pm 0.5$  | 2,358      | 4,522   | 30    |
| ijcnl kernel $L_2$ , $C=1$ | 16  | $5.8 \pm 0.3$  | 1,396      | 4,590   | 98    |

Table 7: Block-size ablation on the synthetic instances (current code): single MDM ( $k=1$ ) versus blocks of  $k$  transfers. Iterations fall with  $k$ ; wall-clock does not fall in proportion, because each block iteration touches more columns.

| instance                                      | $k$ | time (s) | iterations | columns | drops |
|-----------------------------------------------|-----|----------|------------|---------|-------|
| hard margin, $d=1000$ , $\varepsilon=10^{-5}$ | 1   | 0.3      | 296        | 104     | 0     |
| hard margin, $d=1000$ , $\varepsilon=10^{-5}$ | 4   | 0.2      | 36         | 103     | 0     |
| hard margin, $d=1000$ , $\varepsilon=10^{-5}$ | 16  | 0.2      | 16         | 103     | 0     |
| hard margin, $d=1000$ , $\varepsilon=10^{-5}$ | 32  | 0.2      | 16         | 109     | 0     |
| $L_2$ overlap, $d=100$ , $C=1$                | 1   | 10.7     | 75,021     | 1,343   | 0     |
| $L_2$ overlap, $d=100$ , $C=1$                | 4   | 11.2     | 24,606     | 1,632   | 416   |
| $L_2$ overlap, $d=100$ , $C=1$                | 16  | 9.0      | 10,701     | 2,404   | 599   |
| $L_2$ overlap, $d=100$ , $C=1$                | 32  | 10.4     | 7,606      | 3,076   | 305   |
| $L_2$ overlap, $d=1000$ , $C=1$               | 1   | 12.2     | 43,521     | 1,251   | 0     |
| $L_2$ overlap, $d=1000$ , $C=1$               | 4   | 6.7      | 10,071     | 1,330   | 177   |
| $L_2$ overlap, $d=1000$ , $C=1$               | 16  | 4.3      | 3,001      | 1,348   | 180   |
| $L_2$ overlap, $d=1000$ , $C=1$               | 32  | 4.6      | 1,741      | 1,396   | 170   |

**Screening and identification.** Gap-safe rules [14, 16, 17] screen with a radius from the duality gap and identify the support in finite time; the active-set complexity of away-step FW [18] and the strict-complementarity bounds of [19] give the same identification from the algorithm’s side. Theorem 2 is the polytope-distance instance, with constants tied to the TA certificate; the SVM rules of [10, 11] and the exact data reduction of [15] are its ancestors.

**Relation to other FW variants.** The toward, away, pairwise and block rows of the step ablation in Appendix C are Frank–Wolfe, away-step FW, pairwise FW and our block step on the same polytope and objective; Wolfe’s minimum-norm-point method [25] and blended pairwise conditional gradients (BPCG) [12] were not run. BPCG’s blending rule compares the away and FW gaps and its linear rate has the same pyramidal-width form as Theorem 1; our case (b) instead compares the realised gains of the block, MDM, away and toward candidates, so each step is at least as good as the best single candidate, and it is the away fallback rather than the blending that yields the drop-step count. The facial distance of [26] is a computable alternative to the pyramidal width; a bound on  $\delta$  for the difference atom set  $A = \{v_i - w_j\}$  in terms of the two class hulls is open.

**Synthetic results in brief.** Away and pairwise steps converge at  $\varepsilon = 10^{-6}$  where the toward and midpoint steps of [2] do not. Screening reduces 10,000 points to 36–1,054 survivors at  $\varepsilon = 10^{-5}$ . On seven synthetic and small-real problem classes the optimised solver is fastest to a certified solution on six (1.6–2.5 $\times$  ahead of LIBSVM, LIBLINEAR or NuSVC); the exception, MNIST odd-vs-even, is a dense-support  $L_2$  problem. On the four classes where single MDM also converges, blocks and guard add only 0.9–1.5 $\times$  in wall-clock; their clear wins are the two classes (tight tolerance, MNIST) where single MDM does not converge at all.

## E Numerical verification of the analysis

Lemma 1 was checked on 4,000 random ensembles including heavily clipped and boundary cases (worst slack at machine precision); Proposition 1 on 500 near-orthogonal ensembles; Lemma 3

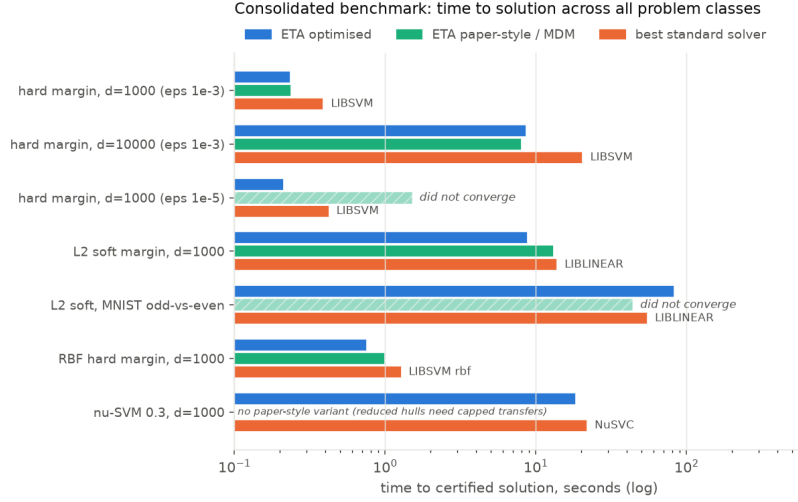


Figure 3: Time to certified solution on seven problem classes: optimised ETA vs the paper-style configuration vs the best standard solver per class. The  $\nu$ -SVM row has no paper-style bar because the toward-step algorithm of [2] has no reduced-hull variant (reduced hulls need capped transfers).

484 to  $10^{-9}$  at arbitrary iterates; Lemma 7 on instrumented two-sided runs (no drop iteration added a  
 485 support index, and every non-drop iteration met the gain bound with  $M$  replaced by the computable  
 486 upper bound  $\text{diam } V + \text{diam } W$ ); and Theorem 2 on instances whose optimal face was computed  
 487 by an exact QP: at every activating round all zero-weight non-face points were screened, and the  
 488 final working set equalled the optimal face exactly. Over 5,000 instrumented live block steps none  
 489 increased the objective. All checks are test files in the released code.